ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** S^{2-}	Z = 16 is sulphur, 2 extra electrons.
3	**(c)** 24.3 u	(1896 + 250 + 286)/100 = 24.32.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** 20 and 16	X lost 2, Y gained 2.
6	**(a)** 20	10x + 11(100 - x) = 1080.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(d)** argon	18 electrons.
9	**(c)** electron	An electron is about 1/1836 of the mass of a proton.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	63x + 65(100 - x) = 6350 (1); x = 75, so the heavier isotope is 25 per cent (1).	1+1
12	Maximum 2n squared with n = 1 is 2 (1); K shell is the smallest and nearest to the nucleus (1).	1+1
13	No dense positive centre in his model (1); It predicted only small deflections (1).	1+1
14	Elements are mixtures of isotopes (1); Atomic mass is the weighted average (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	2, 8, 3 (1); M^{3+} (1); Mass = 54 + 48 = 102 u (1).	1+1+1
16	p + n = 31 and n = p + 5 (1); 2p = 26, so p = 13 and n = 18 (1); Z = 13; configuration 2, 8, 3 (1).	1+1+1
17	Electrons occupy fixed shells (1); No radiation in these states (1); Energy changes only by jumps (1).	1+1+1
	OR: Same valence electrons in a group (1); Across a period valence electrons increase 1 to 8 (1); Chemical behaviour depends on valence electrons (1).	1+1+1
18	Splits on neutron capture giving energy (1); Chain reaction controlled in reactors (1); Radioactive waste and accidents (1).	1+1+1
	OR: Atom had 20 electrons (1); Z = 20 (1); Calcium (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	69x + 71(100 - x) = 6980 (1); x = 60, so 60 per cent and 40 per cent (1); Neutrons: 38 and 40 (1); Element is gallium (1); Average mass between 69 and 71 (1).	1+1+1+1+1
	OR: n = 24 - 12 = 12 (1); B: p = 11, n = 12, A = 23 (1); B is sodium $^{23}_{11}Na$ (1); 2, 8, 1 (1); A is magnesium-24 (1).	1+1+1+1+1
20	Closest approach of alpha particles gives an upper limit for the nucleus size (1); Atom radius about 10^-10 m (1); Density = mass/volume is huge (1); About 10^18 kg per cubic metre (1); Nucleus holds nearly all the mass (1).	1+1+1+1+1
	OR: Li, Na, K have one valence electron (1); Form +1 ions (1); F, Cl, Br have seven valence electrons (1); Form -1 ions (1); Same group, similar properties (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 5	Z = 15: 2, 8, 5.
2	**(c)** 207	Sum.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(c)** Z^{2-}	Gain 2 to make 8.
5	**(a)** tritium	2 : 1.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** hydrogen-1	0 neutrons.
8	**(d)** $^{14}_{6}C$	14 - 6 = 8.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(d)** $^{23}_{11}Na$	Na-23: 23 - 11 = 12.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Z = 9: fluorine (1); $^{19}_{9}F^{-}$ (1).	1+1
12	16 + 4 x 8 = 48 (1); Add 2 for the charge: 50 (1).	1+1
13	Charge +3 (1); 2, 8: resembles neon (1).	1+1
14	Neutrons are uncharged (1); Alpha particles are repelled by the positive nucleus (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(1810 + 6.3 + 202.4)/100 (1); = 20.19 u (1); About 20.2 u (1).	1+1+1
16	Atom: neutral $^{23}_{11}Na$ (1); Ion: Na^{+} has lost an electron (1); Isotope: $^{24}_{11}Na$ has a different number of neutrons (1).	1+1+1
17	Mg (2,8,2) loses 2 electrons (1); S (2,8,6) gains 2 (1); Both reach the octet (1).	1+1+1
	OR: Atom: 11, 12, 11 (1); Ion: 11, 12, 10 (1); Ion is smaller and stable (1).	1+1+1
18	Isobars (1); $^{14}_{6}C$ and $^{14}_{7}N$ (1); Different chemical properties (1).	1+1+1
	OR: (10 x 1 + 11 x 4)/5 (1); = 10.8 u (1); Boron (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	C isotopes differ in neutrons (1); H isotopes: protium, deuterium, tritium (1); Medicine: iodine-131, cobalt-60 (1); Industry: thickness gauges, nuclear fuel (1); Archaeology: carbon-14 dating (1).	1+1+1+1+1
	OR: Na: 2,8,1 reactive metal (1); Cl: 2,8,7 poisonous gas (1); Na^{+} and Cl^{-} stable octets (1); Ionic lattice: hard crystalline solid (1); New substance, new properties (1).	1+1+1+1+1
20	Li 2,1 valency 1 (metal); Be 2,2 valency 2 (1); B 2,3 valency 3 (1); C 2,4 valency 4 (1); N 3, O 2, F 1 (1); Ne 0; trend 1 to 4 and back to 0 (1).	1+1+1+1+1
	OR: H: 1, 0, 1 (1); He: 2, 2, 2; C: 6, 6, 6 (1); O2-: 8, 8, 10 (1); Na+: 11, 12, 10 (1); Cl-: 17, 18, 18 (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** They have different numbers of electrons	Electron numbers are equal.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(c)** 1 km	Ratio 10^5 cm = 1 km.
4	**(c)** 10	11 - 1.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 2, 7, 1	L shell must be full before M.
7	**(a)** neon	Mg2+: 2, 8.
8	**(c)** A is true but R is false.	A is true; R is false: neutrality is due to equal protons and electrons.
9	**(b)** 8	Eight elements fill the L shell.
10	**(c)** 13	10 + 3.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass is 1/1836 u, negligible (1); Its charge balances the protons (1).	1+1
12	Same number of valence electrons (1) (1); Li 2,1; Na 2,8,1; K 2,8,8,1 (1).	1+1
13	H-1: 1 nucleon (1); D: 2 nucleons; T: 3 nucleons (1).	1+1
14	Z = 17 (1); 7 valence electrons (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Alpha particles are helium nuclei (+2) (1); Repelled by the positive nucleus (1); Path bends away (1).	1+1+1
16	Atomic number is unique to each element (1); Isotopes have the same Z but different A (1); Isobars have the same A but different Z (1).	1+1+1
17	Tritium has two extra neutrons (1); Same electron arrangement (1); Chemical properties depend on electrons (1).	1+1+1
	OR: Both have 2, 8, 8 (1); Argon is neutral with a full shell (1); K^+ has a stable octet but is charged and exists in compounds (1).	1+1+1
18	P 2,8,5: 3 (1); S 2,8,6: 2; Cl 2,8,7: 1 (1); Ar 2,8,8: 0 (1).	1+1+1
	OR: Alpha particles are heavy and charged (1); They ionise and lose energy quickly (1); Gamma rays are uncharged and penetrating (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	2, 8, 6: Z = 16 (1); A = 32 (1); Ion S^{2-} (1); Compound Na_2S (1); Mass = 46 + 32 = 78 u (1).	1+1+1+1+1
	OR: Semiconductors from silicon electronics (1); Nuclear power from U-235 (1); Radioisotopes in imaging and therapy (1); Safety: shielding and limited exposure (1); Safe disposal of waste (1).	1+1+1+1+1
20	Electron: Thomson, plum pudding (1); Proton: Rutherford, nucleus (1); Neutron: Chadwick, mass puzzle (1); Bohr: energy shells (1); Models changed with evidence (1).	1+1+1+1+1
	OR: Complete outer shell (1); Very weak forces between atoms (1); Low boiling points (1); Uses: helium in balloons, neon in lights, argon in bulbs (1); Unreactive nature (1).	1+1+1+1+1

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